

Business Data Cleaning, Validation & Excel Reporting

Problem Summary

The project focuses on cleaning and validating retail order data exported from multiple business systems. The objective is to prepare an analysis-ready dataset while documenting all quality issues and generating business reports.

Dataset Description

The dataset contains order-level sales information including customer details, product information, shipping details, payment status, sales, cost, profit, discounts, and order status.

Tools Used

- Microsoft Excel
- Pivot Tables
- Excel Functions (TRIM, PROPER, SUBSTITUTE, IF, YEAR, MONTH, TEXT)
- Conditional Formatting
- Find & Replace
- Flash Fill

Cleaning Steps Performed

- Preserved original dataset.
- Standardized text formatting.
- Removed unnecessary spaces.
- Corrected inconsistent values.
- Standardized date formats.
- Calculated Shipping Delay.
- Identified duplicate records.
- Applied business validation rules.
- Created calculated columns.
- Generated pivot summaries.
- Prepared data quality report.

Business Rules Applied

- Missing Region → Unknown
- Missing Ship Mode → Unknown
- Missing Discount → 0 where appropriate
- Invalid Discounts flagged
- Cancelled Orders excluded from completed sales
- Failed Payments excluded from completed sales
- Refunded Orders reported separately
- Invalid shipping records flagged

Data Quality Issues Found

- Missing values
- Duplicate records
- Invalid discounts
- Date inconsistencies
- Order status issues
- Sales and profit calculation mismatches

Pivot Reports Included

- Sales & Profit by Region
- Sales & Profit by Category and Sub-category
- Order Count by Ship Mode
- Profit Margin by Customer Segment
- Refunded, Cancelled and Failed Orders by Region
- Monthly Sales Trend

Key Business Insights

- Regional sales performance can be compared easily.
- Product categories contributing the highest sales are identified.
- Customer segment profitability is highlighted.
- Shipping performance can be monitored through delay analysis.
- Cancelled and refunded orders are separately tracked for operational review.

Assumptions

- Missing discounts treated as zero only when business rules allowed.
- Invalid records retained and flagged instead of being deleted.
- Duplicate Order IDs with conflicting values require manual review.

Limitations

- Manual review is required for conflicting duplicate records.
- Some business assumptions depend on source system accuracy.

Repository Structure

studentname_studentid_part1_data_cleaning/

- data/
- outputs/
- screenshots/
- README.md

Screenshots Included

- raw_data_preview.png
- cleaned_data_preview.png

- pivot_summary_1.png
- pivot_summary_2.png